

Two-frame weighted PCA precleaning of near-infrared spectra, and what it does to line-by-line velocities

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Abstract

Near-infrared radial velocities are limited less by photon noise than by what is left of the atmosphere and of the instrument in a corrected spectrum. We model a campaign with two rest frames at once: one anchored on the star, one anchored on the observer, fitted together on a log-uniform grid where a Doppler shift is an exact translation. Dividing the observer block out of every exposure leaves a spectrum whose stellar content is untouched, which is then measured line by line. On a SPIRou campaign of TOI-2120, where telluric residuals dominate, the velocity scatter falls from 47.7 to 20.4 m s^{-1} . On three NIRPS campaigns whose delivered velocities are already good, the same correction is neutral at best. A pair of twin experiments, two 14-night slices of one campaign at the same signal-to-noise, shows what decides the outcome: where the barycentric velocity spans 42.8 km s^{-1} the correction improves the robust scatter from 8.5 to 4.5 m s^{-1} , and where it spans 0.7 km s^{-1} it degrades it from 9.1 to 21.8 . When the star does not move against the sky, the two frames are one frame, the observer block takes up stellar structure, and dividing it out removes part of the star. We give a practical guard, and a mode that fits several stars against a single observer basis, which breaks the degeneracy by construction.

1 Introduction

In the near infrared the error budget of a precise radial velocity is rarely photon noise. Telluric absorption and emission, detector systematics and the imperfect removal of both leave a residual that is fixed in the observer’s frame while the star moves through it at the barycentric velocity. Telluric correction removes most of it; what remains is small in flux and large in velocity.

The line-by-line method of Artigau et al. (2022) measures a velocity per line and combines them with an outlier-resistant estimator, which is what makes a residual of this kind visible: a handful of contaminated lines no longer drags the whole exposure. It also makes it worth removing the residual before the velocities are measured rather than after.

This paper describes a data-driven precleaning of that residual, and, as much as the method itself, what four campaigns say about when it helps. The implementation, the configuration of every run shown here and the tests that pin its behaviour are in one repository; each figure and each number below is produced by the pipeline it documents.

2 Method

2.1 Two frames, one model

Each exposure is resampled onto a grid uniform in $\ln \lambda$, where a Doppler shift is a translation of $\text{atanh}(v/c)$ divided by the grid's step, exactly, with no approximation in v/c . A Savitzky-Golay filter of 100 km s^{-1} removes the continuum, so the data are $y = \ln f - \text{savgol}(\ln f)$, in which an absorption of optical depth τ is $-\tau$ and scales linearly with the column it comes from.

A t.fits exposure gives two rows, the even orders and the odd ones, which see the same lines at different resolutions. The model of row n is

$$y_n = T_{p(n)} + a_n S_n P + b_n \cdot Q + \epsilon_n, \quad (1)$$

where T_p is the star's own spectrum for order parity p , carried into the exposure's frame by S_n , the shift operator of that exposure's barycentric velocity; P are components anchored in the star's rest frame; Q are components anchored in the observer's; and a_n , b_n are that exposure's amplitudes, shared by its two parity rows because they are one measurement of one star at one instant.

Two properties of Eq. 1 matter for what follows.

The star's mean spectrum carries the coefficient one. T_p is estimated once, before any component, as the barycentric-binned median of the rows of that parity in the star's frame, and is then subtracted with a coefficient of exactly one. A coefficient in front of a term in a logarithm is an exponent on the flux, and an exponent on a star's mean spectrum describes no star. The free amplitude a_n multiplies P , which describes variability around that fixed spectrum, and is measured below to be nearly orthogonal to it.

A free amplitude on an observer component is exact physics. For an absorption, $\ln T = -\tau$ and the optical depth scales with the airmass and the column, so b_n is the physical parameter itself, not a linearisation.

2.2 The star per order parity

The two halves of the echellogram are not equivalent. Where two orders overlap, the same stellar lines appear at both ends of a blaze, and their measured depths differ: on Proxima at 1201.1 nm the four lines of one overlap are shallower in the odd orders by 0.005 to 0.042 in $\ln f$, and the even and odd residuals of a single fit are mirror images of one another, with a rank correlation of -0.98 . One star spectrum cannot describe both halves. The model therefore carries one T_p per parity, and only the amplitudes are shared.

2.3 Weights, and the samples nobody should trust

Every sample carries a weight: photon noise, floored by the sample-to-sample scatter measured over 201 samples, which is what keeps the thermal background of the reddest orders from being called the quietest part of the detector; a ramp in telluric transmission; zero below a line depth of $\ln f = -0.5$; zero at the edges of the grid; and zero wherever the Lanczos taps of the carry would reach into a hole in the basis support.

2.4 Solving

The two blocks are solved by block coordinate descent. Each sweep solves the amplitudes of both blocks jointly per exposure, by the normal equations of the design $[S_n P^\top \mid Q^\top]$, the two parity rows summed into one system; then updates each basis against the residual of the other. The star side is a cubic B-spline with a knot per sample, evaluated at each exposure’s shifted positions and updated exactly by a banded solve, which is both faster and slightly better conditioned than carrying samples with a Lanczos kernel.

2.5 The correction

Only the observer block is divided out:

$$f_{\text{corrected}} = f \exp(-b_n \cdot \tilde{Q}), \quad (2)$$

on each order’s own pixels, by a cubic spline from the grid. The star block is never divided out: with no external template the first star component is the star.

$\tilde{Q}$ is the fitted basis shrunk by its own significance, column by column. At column i the components’ Fisher matrix is $F_i = \sum_n w_{ni} b_n b_n^\top$, with the weights scaled by the fit’s median reduced χ^2 ; $\sigma^2(Q_{ji}) = [F_i^{-1}]_{jj}$ and $z_{ji} = Q_{ji}/\sigma(Q_{ji})$; and

$$\tilde{Q}_{ji} = \max(0, 1 - z_{ji}^{-2}) Q_{ji}. \quad (3)$$

The significance is the basis’s, measured with every exposure at once, not one exposure’s share of it: a pattern at half a sigma in each of N exposures is detected at about $0.5\sqrt{N}$ and is kept, while a column where the component is noise is divided out at zero and the file keeps its delivered flux there.

Finally, a sample the fit gave no weight cannot be corrected, and is blanked. Blanking each exposure’s own leaves every epoch with a different set of lines; we show below that this alone costs 1.8 ms^{-1} on one campaign. The nominal mode blanks, in every exposure, every sample that any exposure lost, so the campaign carries one set of lines.

3 Data

Four campaigns, two instruments, reduced with APERO (Cook et al., 2022) and measured with LBL (Artigau et al., 2022).

Target	Instrument	T_{eff} (K)	exposures	nights	delivered rms (m s^{-1})
TOI-2120	SPIRou	3179	321	—	47.7
Proxima	NIRPS	3050	782	259	2.9
GJ 1	NIRPS	3478	292	146	2.7
TOI-4552	NIRPS	3304	119	60	12.6

Table 1: The campaigns. Nights are the nightly coadds the NIRPS fits use; the delivered rms is the line-by-line velocity scatter of the spectra as reduced, with no precleaning.

Target	delivered		corrected	
	rms	robust σ	rms	robust σ
TOI-2120	47.71	34.99	20.35	16.16
Proxima	2.86	2.22	3.04	2.52
GJ 1	2.70	2.74	2.85	2.78
TOI-4552	12.64	11.37	15.35	14.56

Table 2: Line-by-line velocities, in m s^{-1} , on the exposures both series share. The correction transforms the campaign whose residuals are telluric, and costs where the delivered velocities are already good.

4 Results

4.1 What the correction is worth

The pattern of Table 2 is not a signal-to-noise effect: GJ 1 is four magnitudes brighter than TOI-4552 and both are NIRPS campaigns stacked by night, yet one is neutral and the other loses 3 m s^{-1} .

4.2 The barycentric span decides

TOI-4552 was observed in two seasons. In the first the barycentric velocity spans 52.6 km s^{-1} ; in the second, 9.8. Fitting each season alone, the correction improves the first and degrades the second. To remove every other difference we then cut two slices of 14 nights each, at the same median signal-to-noise, one spanning 42.8 km s^{-1} of barycentric velocity and one spanning 0.7 (Fig. 1):

14 nights of TOI-4552	delivered		corrected	
	rms	robust σ	rms	robust σ
BERV span 42.8 km s^{-1}	7.80	8.48	6.18	4.52
BERV span 0.7 km s^{-1}	8.96	9.14	21.76	21.76

Table 3: The same campaign, the same instrument, the same number of nights and the same signal-to-noise. Only the barycentric coverage differs, by a factor of sixty, and the correction changes sign.

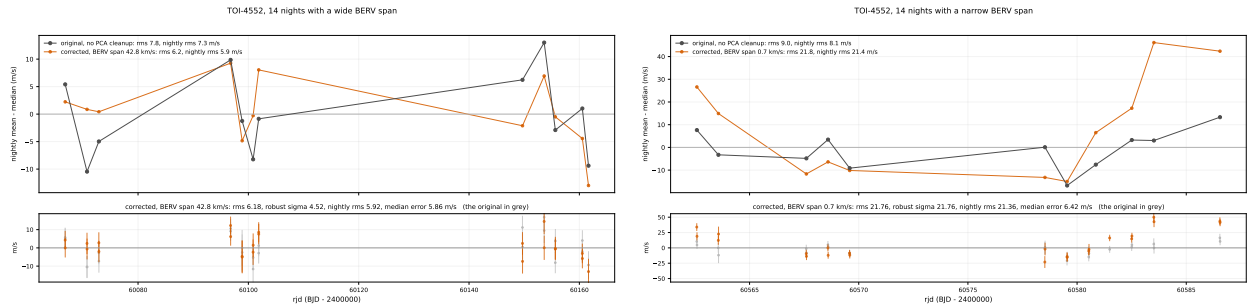


Figure 1: Left: the wide-span slice, corrected in orange over the delivered series in grey. Right: the narrow-span slice. Nightly means above, exposures below.

4.3 What does not explain it

Every other candidate was measured on the same 109 exposures of TOI-4552 and ruled out: nightly stacking (fitting the exposures individually gives 15.6 against 15.9 ms^{-1}); the number of observer components, in the fit (16.9 with one instead of three) or in the correction (15.9 with one divided out); the velocity term (16.4); the shrinkage, which helps (15.1 with it, 18.1 without); the template handed to LBL; and the mask.

4.4 The mask, and the set of lines

Blanking the samples the fit could not weight is necessary, but blanking a different set in every exposure is not. With nothing divided out at all and each exposure blanking its own, the scatter rises from 12.64 to 14.41 ms^{-1} while the per-exposure error does not move: a line set that moves from epoch to epoch is scatter, not lost information. Blanking one common set instead gives 11.96, below the delivered spectra themselves, for 4.8 percentage points more of each exposure blanked.

4.5 The star’s freedom, and what a_1 is

The amplitude a_1 multiplies the star-frame variability component, never the star’s mean spectrum. Measured on four campaigns, that component is nearly orthogonal to the mean spectrum, $|\cos| \leq 0.065$; expressed as an exponent on the star it amounts to 0.4% rms on the three NIRPS targets and 1.25% on TOI-2120.

It is nevertheless not describing the star. On Proxima, a_1 follows the barycentric velocity with a rank correlation of 0.69, ahead of humidity (0.43) and airmass (-0.39), and its periodogram peaks at 366 days (Fig. 2). It is an annual mode aligned with the Earth’s orbit.

Removing the star component altogether, so that the star is its fixed template and nothing else, takes TOI-4552 from 14.56 to 11.20 ms^{-1} of robust scatter at equal mask and template (Fig. 3), that is, below the delivered spectra. The freedom the star block is given is used, in part, to absorb what belongs to the observer, and the observer block then divides out what belongs to the star.

5 Discussion

5.1 One frame instead of two

The observer block is identifiable only because the star moves through it. Over a campaign whose barycentric velocity spans tens of km s^{-1} , a pattern fixed in the observer’s frame lands on different stellar lines in every exposure, and no combination of stellar structure can imitate it. Over a campaign whose barycentric velocity barely moves, the two frames coincide: a column of the observer basis and a column of the star’s spectrum are then the same column, and the fit has no way to tell which of them owns the flux. Dividing the observer block out then removes part of the star, and the line-by-line velocities record it as a barycentric-shaped injection.

This is the same mechanism, at the campaign scale, as the annual mode that a_1 carries.

5.2 A guard, and a way out

The practical guard is to look at the barycentric span of the set of exposures a fit is built on, not merely at their number, and to fit seasons separately when a campaign mixes wide and narrow spans:

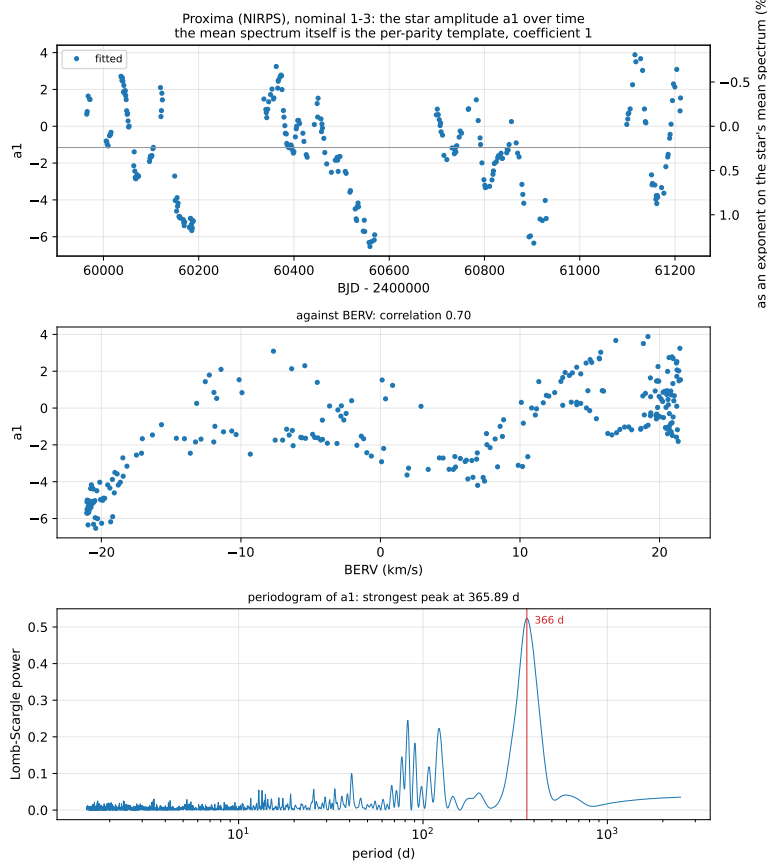


Figure 2: Proxima: the star amplitude over time, against the barycentric velocity, and its periodogram. The right-hand axis of the first panel gives the equivalent exponent on the star’s mean spectrum.

on TOI-4552, fitting the narrow season alone costs 14% against the delivered velocities, where fitting it together with the wide one costs 55%.

The principled way out is to break the degeneracy with other stars. The atmosphere and the instrument belong to the night, not to the target, while the star, its barycentric coverage and its systemic velocity do not: several campaigns fitted against a single observer basis cannot let that basis follow any one star. The pipeline has a mode for it, in which the rows of several objects share one grid and one observer basis while each object keeps its own star spectrum per order parity.

5.3 What the precleaning is for

Table 2 is the honest summary: this is a method for campaigns whose velocities are limited by what the atmosphere and the instrument left behind, and it is not free where they are not. On TOI-2120 it removes more than half the scatter. On Proxima and GJ 1 the delivered velocities are already at 2.7 to 2.9 ms^{-1} , and the correction adds 0.2. Applying it there buys nothing, and the decision can be made in advance from the barycentric span and from how much of the scatter the formal errors already explain.

TOI-4552: what the star's freedom costs, at equal mask and template

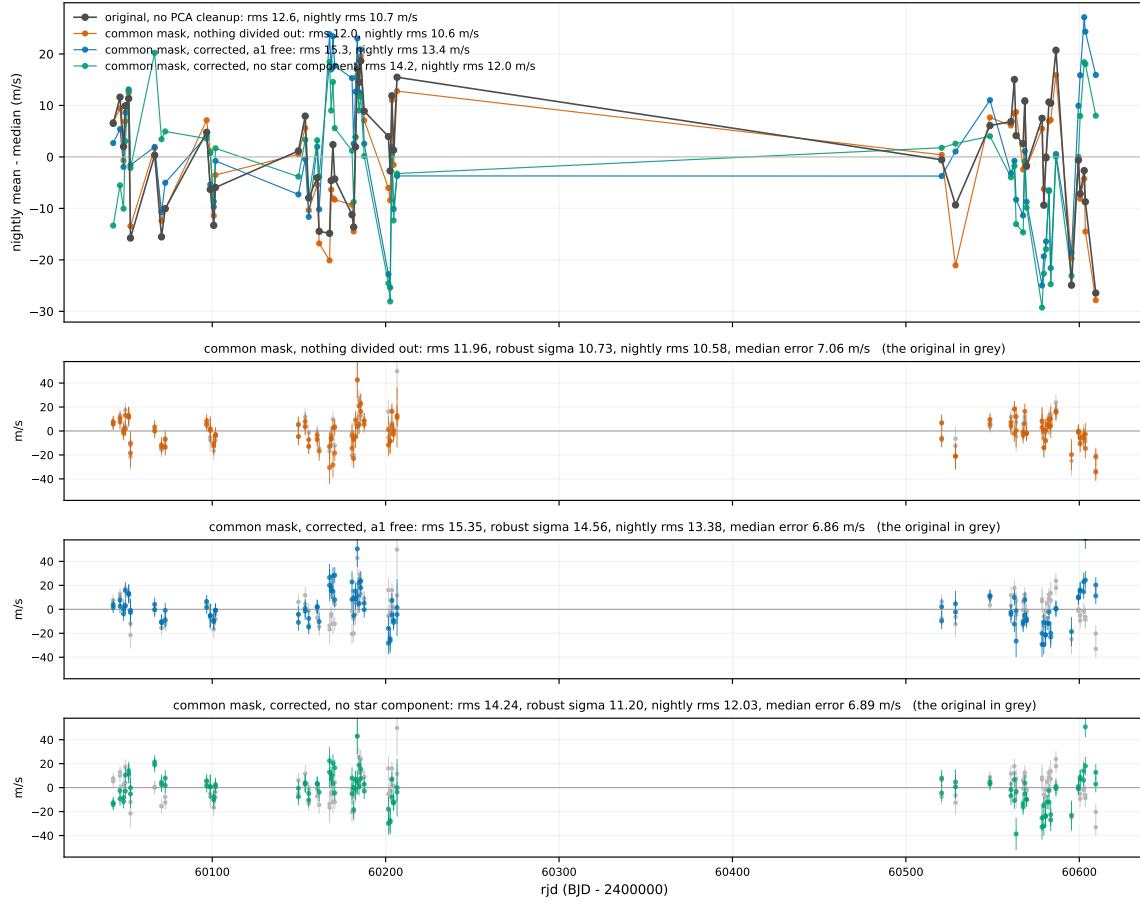


Figure 3: TOI-4552 at equal mask and template: the delivered series, the same spectra with the mask alone, the correction with a free star amplitude, and the correction with no star component at all.

6 Conclusions

1. A two-frame weighted decomposition separates what is fixed in the observer's frame from what belongs to the star, and dividing the first out removes more than half of the velocity scatter of a campaign dominated by telluric residuals.
2. What decides whether it helps is the barycentric span of the fitted set. Two 14-night slices of one campaign, at equal signal-to-noise, differ by a factor of sixty in span, and the correction improves the robust scatter by a factor of two in one and degrades it by a factor of 2.4 in the other.
3. The star's mean spectrum must carry a coefficient of exactly one. The freedom given to a star component is used to absorb an annual, barycentric mode, and removing it recovers most of the damage on the worst case.
4. Blanking the samples a fit could not weight must be done with one set of lines for the whole campaign. Blanking each exposure's own costs 1.8 ms^{-1} with no correction applied at all.

References

- Artigau, É., et al. 2022, *Line-by-line Velocity Measurements: an Outlier-resistant Method for Precision Velocimetry*, AJ, 164, 84, doi:10.3847/1538-3881/ac7ce6
- Cook, N. J., et al. 2022, *APER0: A PipelinE to Reduce Observations, Demonstration with SPIRou*, PASP, 134, 114509, doi:10.1088/1538-3873/ac9e74